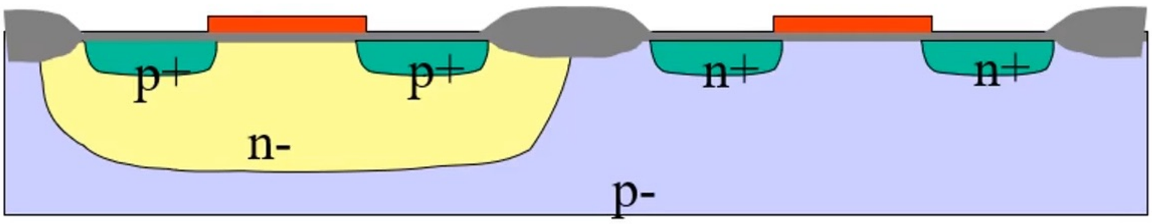
Lecture 1

→ Semiconductors & MOS-Transistors

Sideview of Silicon:

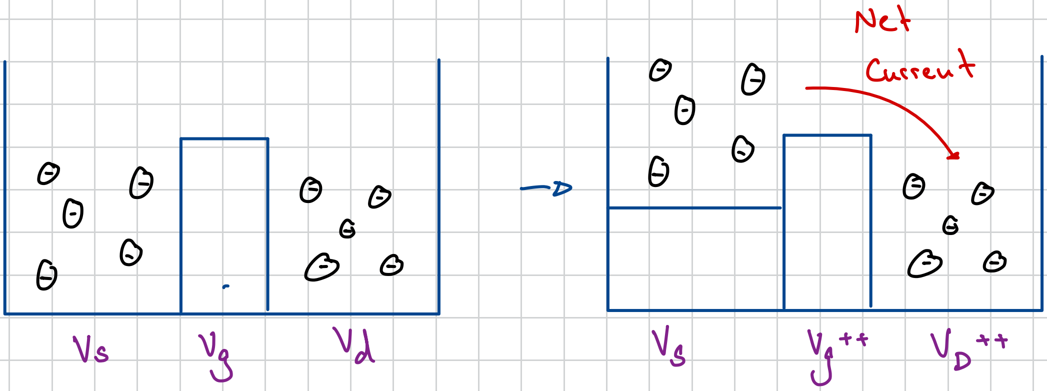
→ Gate controls height of energy barrier between Source & Drain

Source Gate Drain



p-type transistor

n-type transistor

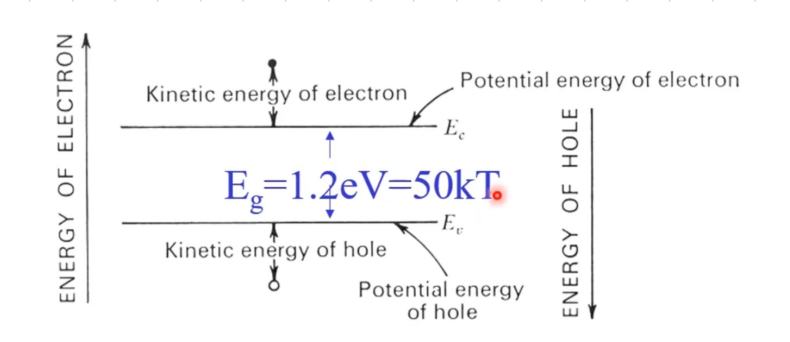


→ Donors : Elments with 4< electrons in outermost shell

→ Acceptors : Elmts with 4> electrons in outermost shell

→ Hole : Missing Electron → Quantum Mechanicaly behaves like a Particle

↳ Can move and so on like an electron
↳ Have positive Charge
↳ 2.5 times the effective charge of an electron



eV = electron Volts

k = Boltzmann Konstant

↳ $1.38 \cdot 10^{-23} \frac{J}{K}$

T = absolut Temperatur

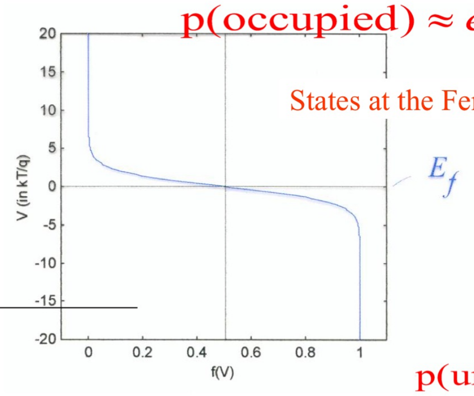
The Fermi-Dirac Distribution

States above Fermi level are **occupied** with Boltzmann distribution

$$\text{p(occupied)} \approx e^{-(E-E_f)/kT}$$

→ describes distribution of Holes and electrons

Energy relative to Fermi level in kT units



States at the Fermi level are ½ occupied

States below Fermi level are **unoccupied** with Boltzmann distribution

$$\text{p(unoccupied)} \approx e^{-(E_f-E)/kT}$$

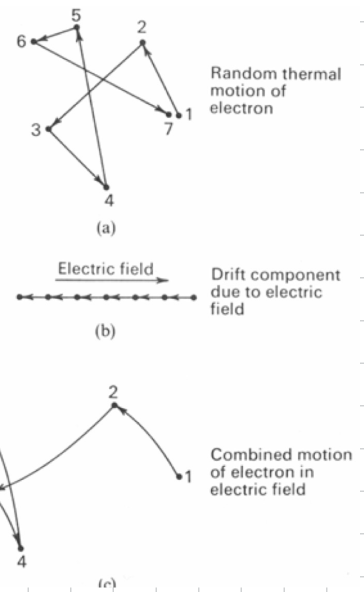
Probability of occupation of a state

→ Law of mass action : $np = n_i^2$

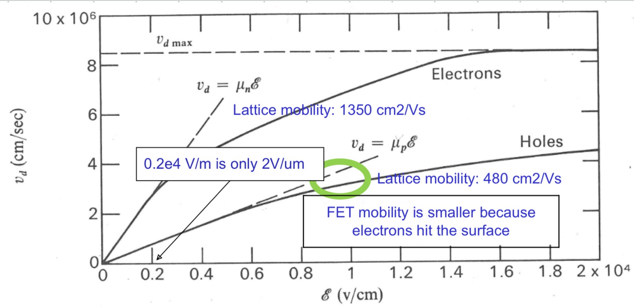
↳ the more holes, the less e^- and vise vosa

Electron Movement :

① Drift :



→ Mobility is a function of the electric field



→ Random motion due to temp + electric field result in drift

Velocity

$$\vec{J} = qnv = qn\mu(\vec{\xi})$$

Electric field

Mobility

Current Flux

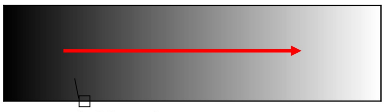
Charge density

② Diffusion

→ function of concentration density

→ Drift & diffusion are related by Einstein Relation

Gray scale shows density of carriers that are moving at avg thermal velocity of ~100km/s



Diffusion current

$$\vec{J} = -Dq\vec{\nabla} n$$

Current Flux

Diffusion constant

Spatial gradient of charge density

Drift and diffusion are related by the *Einstein Relation*

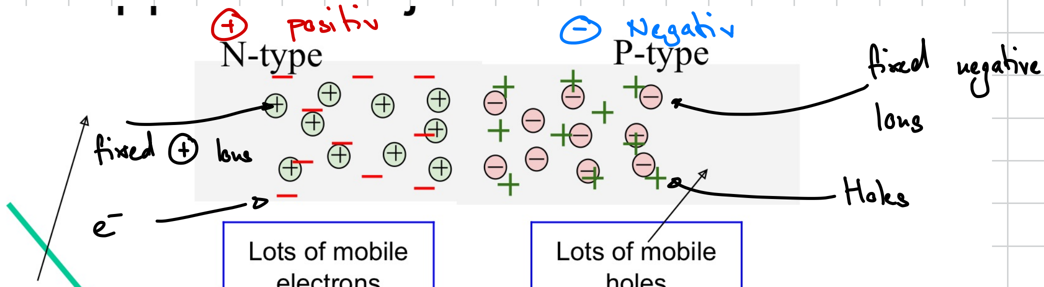
Jdiffusion = -DqΔ n

Jdrift = μq n ξ

$$D = \frac{kT}{q}\mu$$

D is proportional to absolute temperature *T*

Charges, fields, potentials in PN Junctions



Lots of mobile electrons

Lots of mobile holes

Electron density gradient: Electrons diffuse rightwards

Hole density gradient: Holes diffuse leftwards

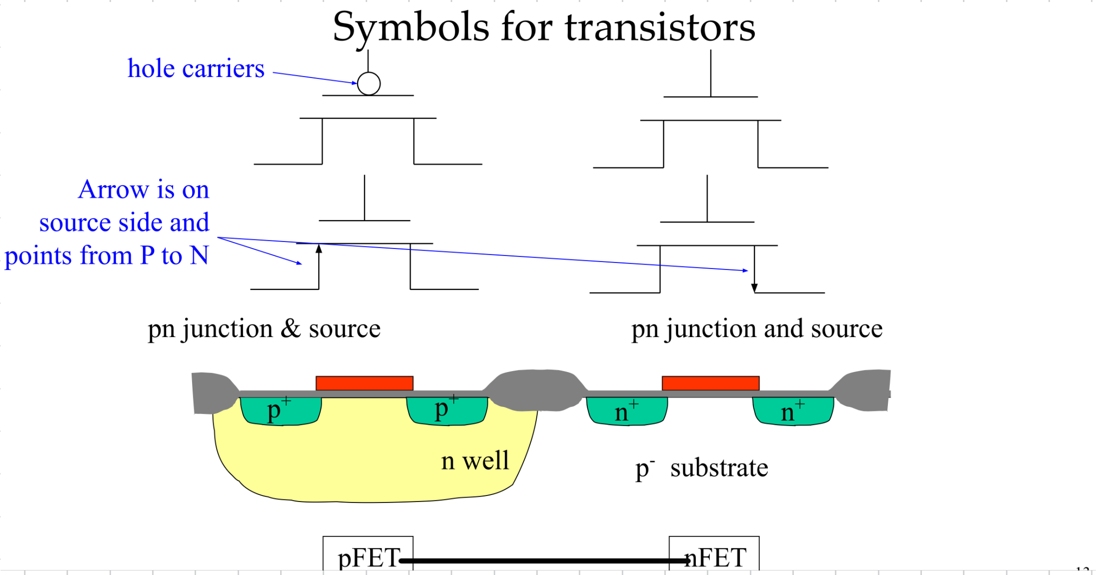
The charge separation builds up an electric field to pull the carries "back home"

In a PN junction with no applied voltage, drift and diffusion are balanced everywhere

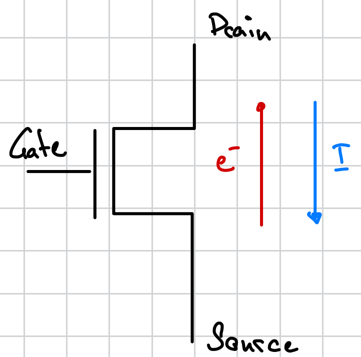
e⁻ diffusion →

← holes diffussion

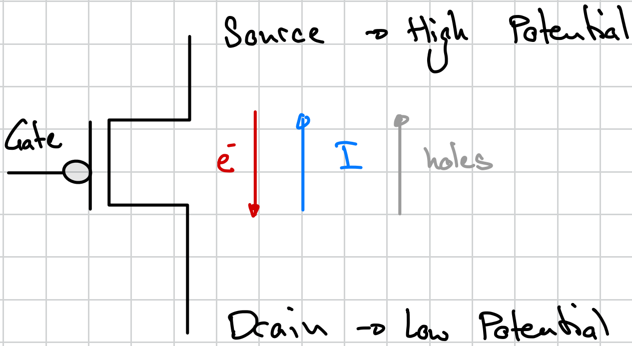
Lecture 2



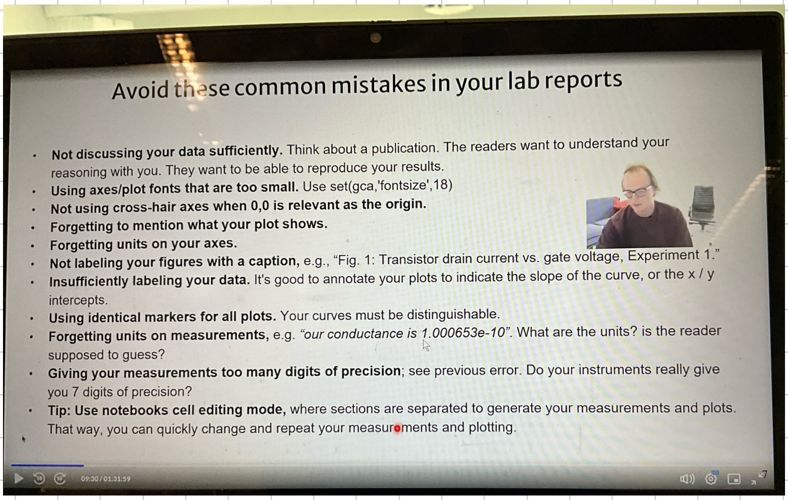
NMOS : → Carries e^-



PMOS : → carries Holes



→ Mistakes in Jupyter NB :



Constants :

q = charge of a electron

$U_T = kT/q$ → Thermal voltage $\approx 26\ mV$

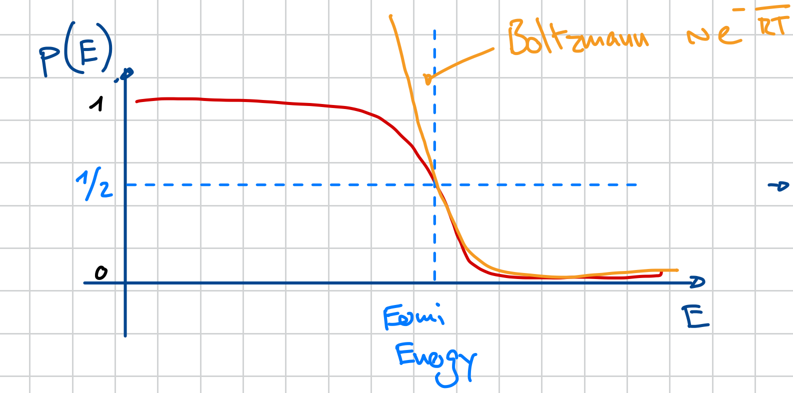
ε = Permittivity of vacuum

$$\text{Electric Field} = \frac{\overset{\frac{Q}{A}}{\sigma}}{2\varepsilon_0\,\varepsilon}$$

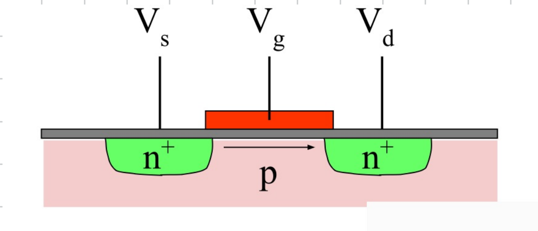
ε_0: Permittivity of vacuum

ε: relativ Permittivity

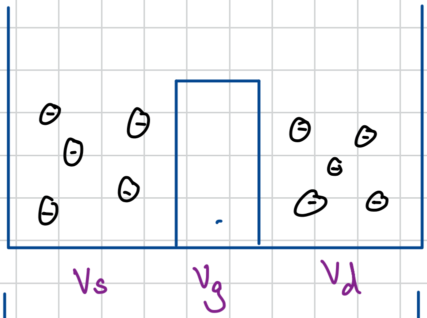
→ Fermi - Dirac Distribution:



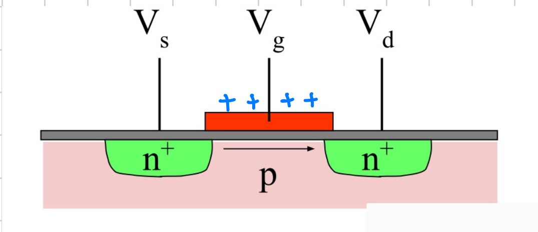
→ In areas of low Energy, its very likely that an e⁻ appears



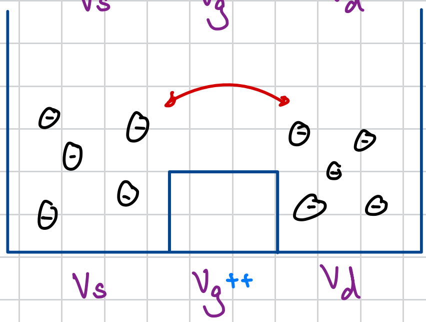
→



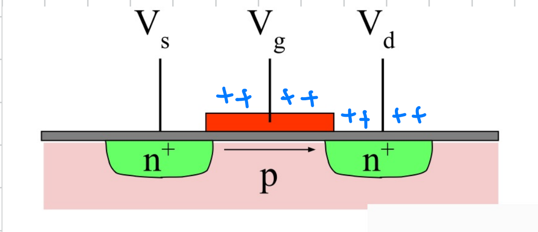
→ No Movement



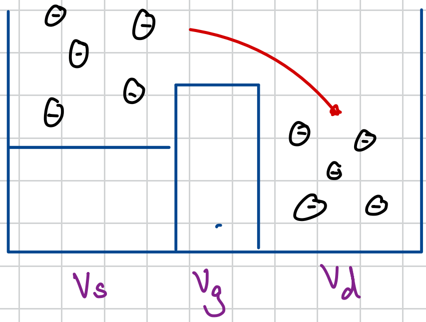
→



→ Movement but no net - Flow



→



→ Net - flow from Source to drain

→ Lower Energy to higher probability that electrons are there

→ Similarity's between Neurons & Transistors

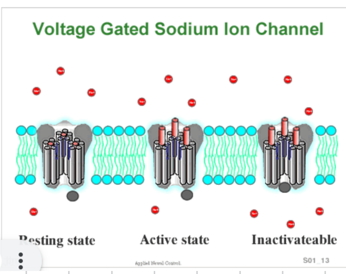
Neuron channels and Transistors

Both depend on exponential Boltzmann distributions of Concentration vs. Energy.

Neurons

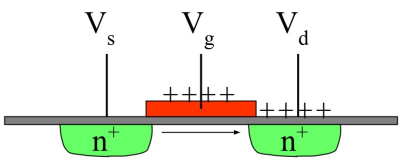
→ needs much less voltage to activate

- **Membrane ionic conductance** is exponentially dependent on the **voltage across the neuron membrane**.
- The **population of open channels** depends exponentially on potential across barrier.



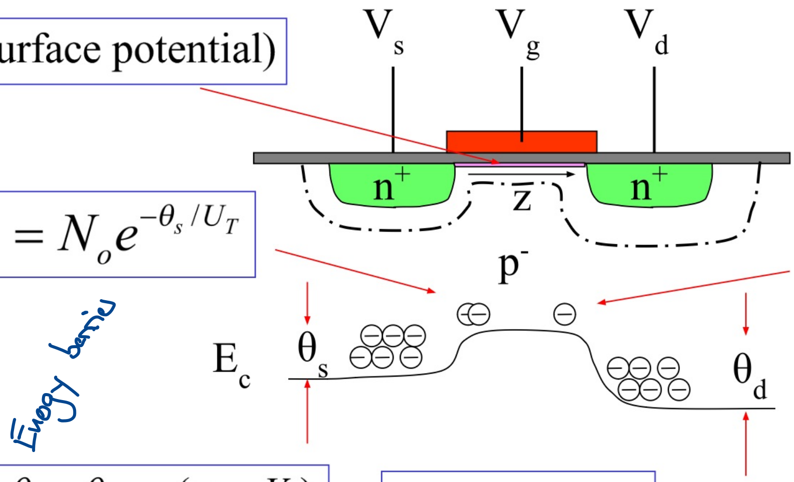
Transistors

- **Current flow** in transistors is exponentially dependent on **barrier height**.
- The **population of carriers** depends exponentially on the barrier height.



Subthreshold nFET: Current is diffusion current

ψ_s (surface potential)



$$N_s = N_o e^{-\theta_s / U_T}$$

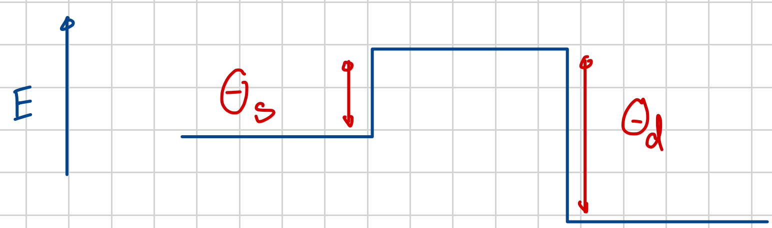
$$N_d = N_o e^{-\theta_d / U_T}$$

$$\theta_s = \theta_0 - q(\psi_s - V_s)$$
$$\theta_d = \theta_0 - q(\psi_s - V_d)$$

$$\frac{dN}{dz} = \frac{N_s - N_d}{L}$$

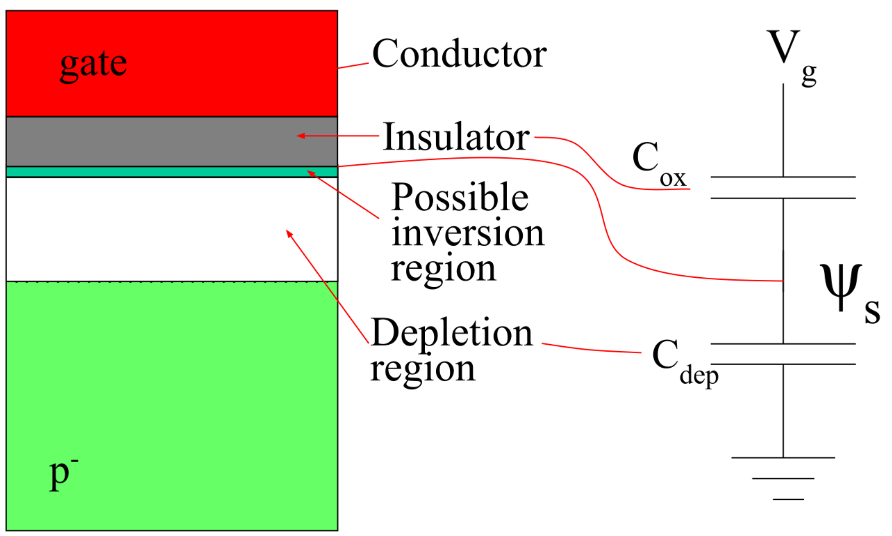
$$I = -qWD_n \frac{dN}{dz} = I_0 e^{\psi_s / U_T} \left(e^{-V_d / U_T} - e^{-V_s / U_T}\right)$$

N=carrier density per unit volume
W=channel width
L=channel length
D=diffusion constant
θ_o=built-in potential barrier
N_s=built in concentration
I_0=subthreshold leakage current



$V_s = V_0 - q\Psi_s + qV_s$

Influence of gate on surface potential ψ_s



ψ_s is controlled only by V_g and the bulk voltage (ground for n-fet), through the capacitive divider C_{ox} and C_{dep}

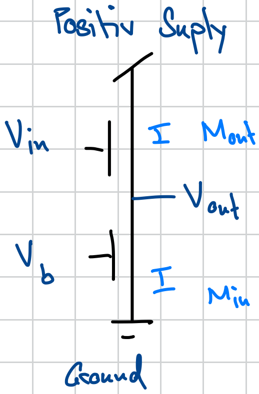
Ψ_s = *Surface potential*

$$\kappa(kappa) = \frac{\partial \psi_s}{\partial V_g} = \frac{C_{ox}}{C_{ox} + C_{dep}}$$

$\rightarrow$ Kappa = $\dfrac{\text{rate change of Surface Potential}}{\text{Gate Voltage}}$

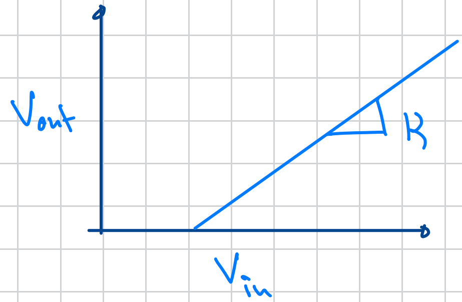
Puzzle Lecture ②:

→ Circuit calle "Source Followo"



$$e^{KV_b} = e^{KV_{in} - V_{out}}$$

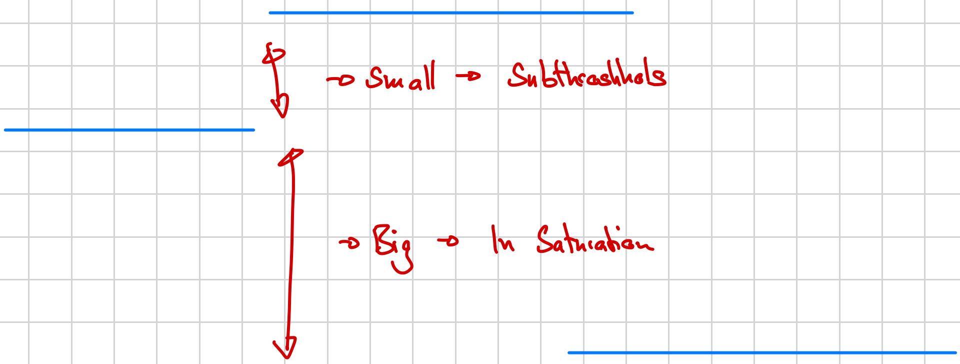
$$V_{out} = K(V_{in} - V_b)$$



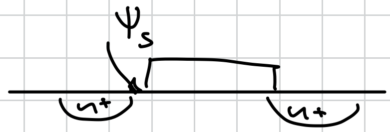
$$K = \frac{C_{ox}}{C_{ox} + C_{dep}}$$

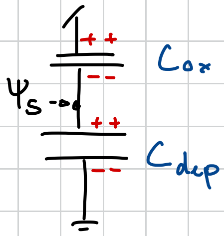
Saturation bezieht sich auf V_{DS} – Value

Threshhold bezieht sich auf: V_{GS} – Value



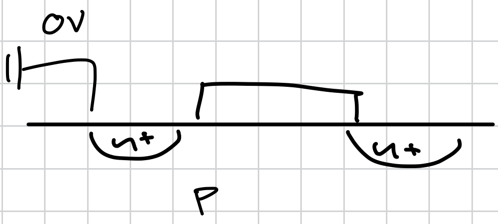
Lecture 3





$$Q = C_{ox}(V_g - \Psi_s) = C_{dep}\,\Psi_s$$

$$\Psi_s = V_g \frac{C_{ox}}{C_{ox} + C_{dep}}$$



accumulation : $V_g < 0$

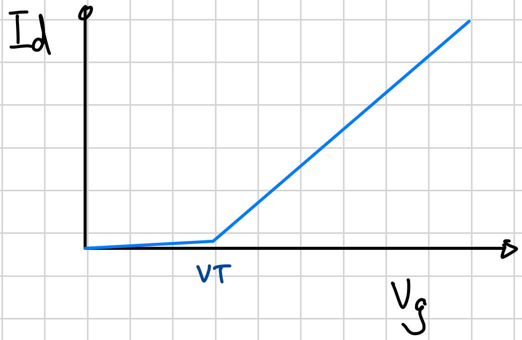
Weak Inversion / depletion : $0 < V_g < V_T$

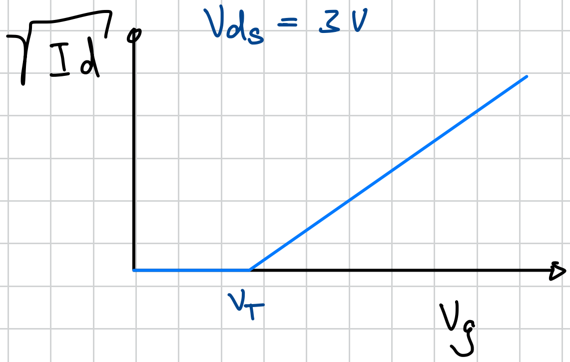
Strong Inversion : $V_g > V_T$

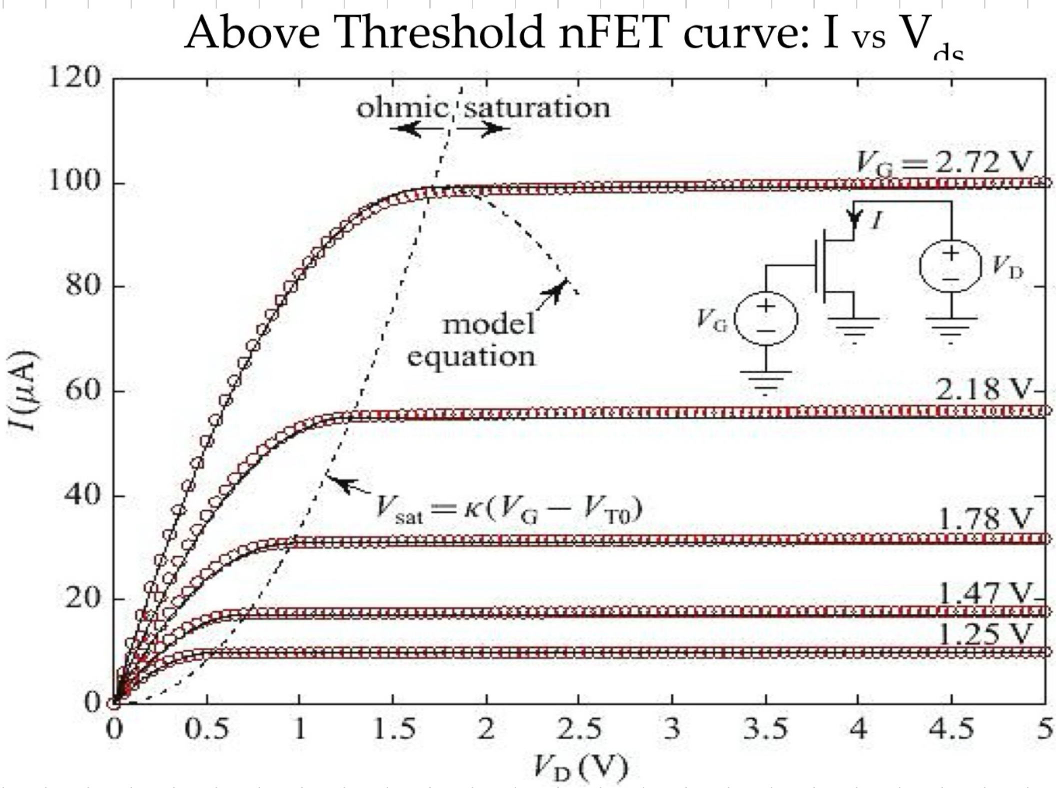
→ small V_{ds} $\quad I_{ds\ linear} = \mu E n = \mu \frac{V_{ds}}{L} \underbrace{(V_{gs} - V_T)}_{\text{Overdrive}} C_{ox} W$

Overdrive → how much V_{gs} is over threshold

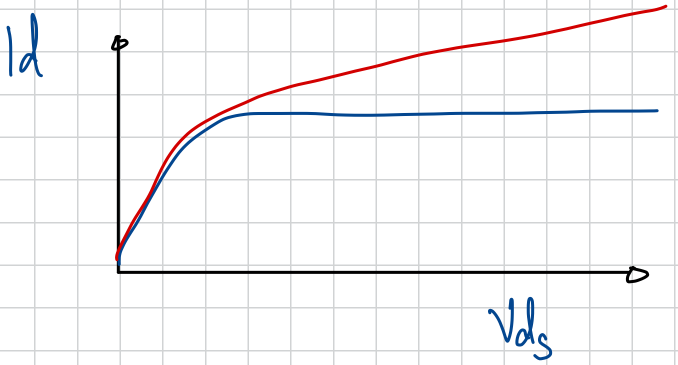
$$= \underbrace{\mu C_{ox} \frac{W}{L}}_{\beta} V_{ov} V_{ds} \qquad V_{ov}$$

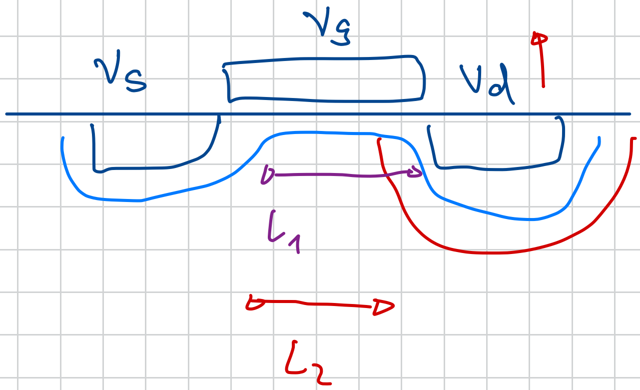






→ the higher the gate voltage, the higher the current at saturation



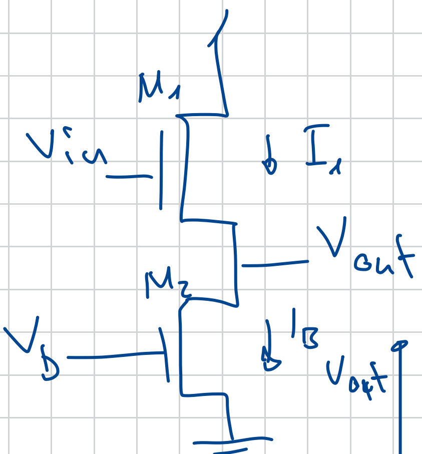


→ Channel Length modulation with rising Vd

→ "Early" effect

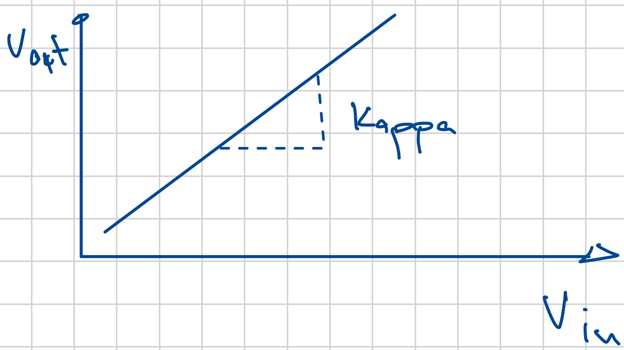
↳ Smaller "Shorter" Transistors are more effected

Source Follower

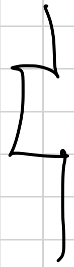


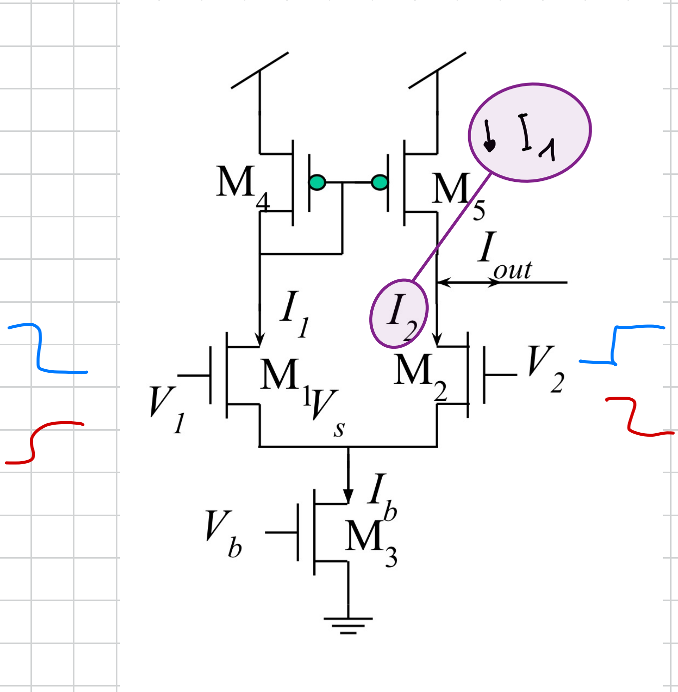
$$I_1 = I_B$$

$$V_{out} = k(V_{in} - V_b)$$



C2F : $f = \frac{I}{C \Delta U}$



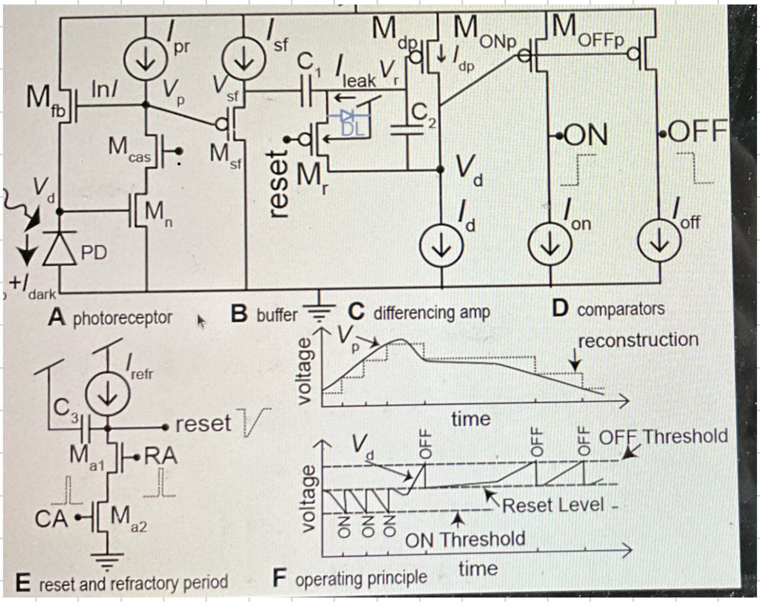


$I_1 = \frac{I_b}{2} - \varepsilon I$ $\qquad$ $I_2 = \frac{I_b}{2} + \varepsilon I$

$I_1 = \frac{I_b}{2} + \varepsilon I$ $\qquad$ $I_2 = \frac{I_b}{2} - \varepsilon I$

→ $I_1 \neq I_2$ what means we have a change in the V_s of M_2

DVS Lab



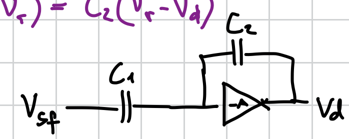
$$\to C_1 V_{sf} = \overbrace{\frac{-C_1 - C_2}{A}}^{\Rightarrow 0} V_d - C_2 V_d$$

$$V_d = -\frac{C_1}{C_2} V_{sf}$$

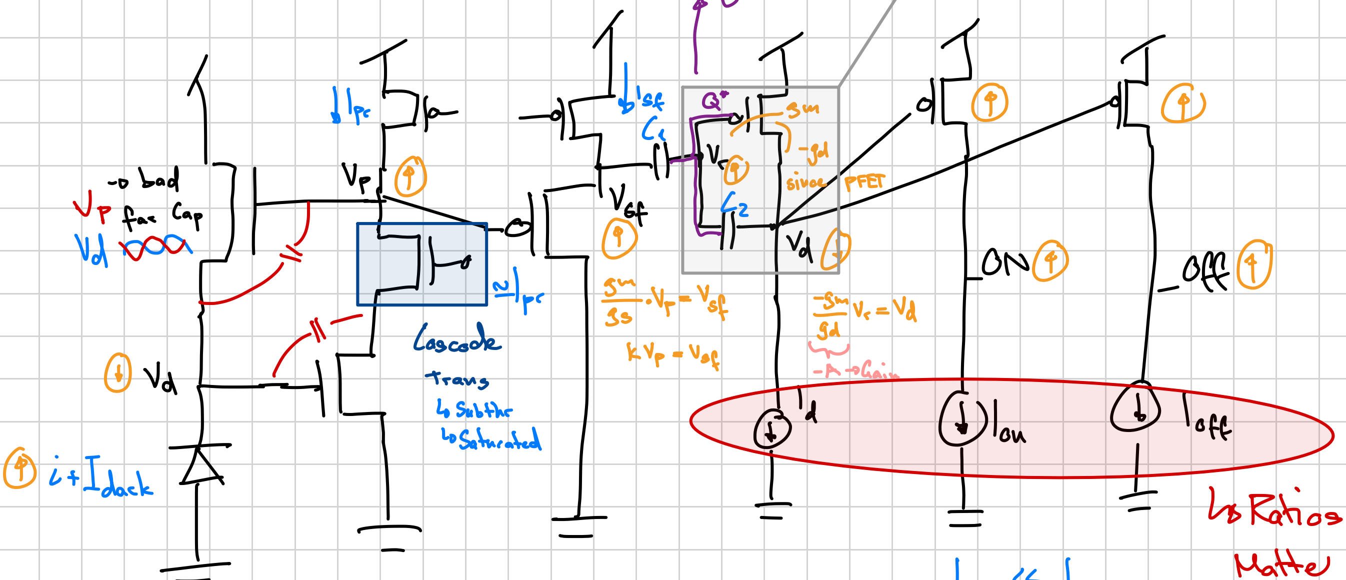
$$\to C_1\left(V_{sf} + \frac{V_d}{A}\right) = C_2\left(-\frac{V_d}{A} - V_d\right)$$

Charge

$$\to Q^* = C_1(V_{sf} - V_r) = C_2(V_r - V_d)$$



In theory this charge is constant → Caps act as Memory

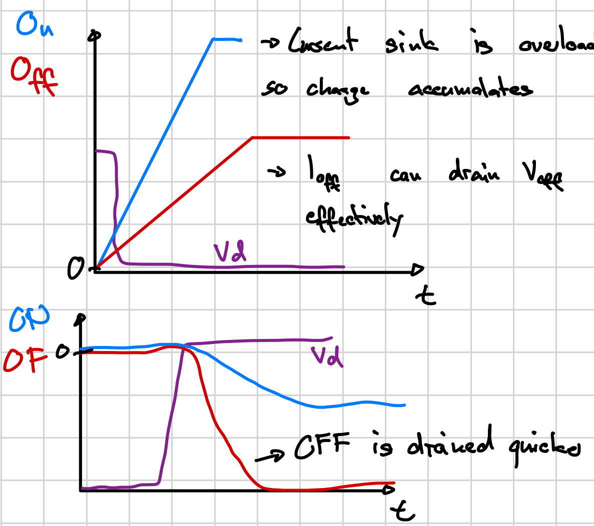


↳ Ratios Matter

$I_{on} \ll I_{off}$

→ Miller Caps: → Parasitic Capacitance

↳ Every trans is a bit of a Cap → Miller Capacitance is importa since it is between two nodes that move apart



→ Current sink is overloaded so charge accumulates

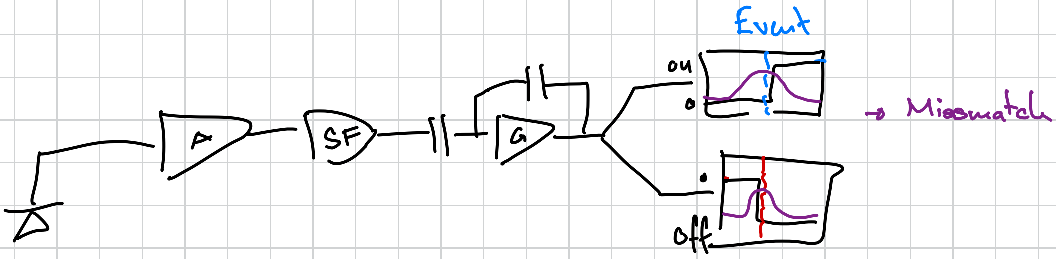
→ I_{off} can drain V_{off} effectively

→ OFF is drained quicker

→ Cascode:

→ We have cap between two nodes that move apart

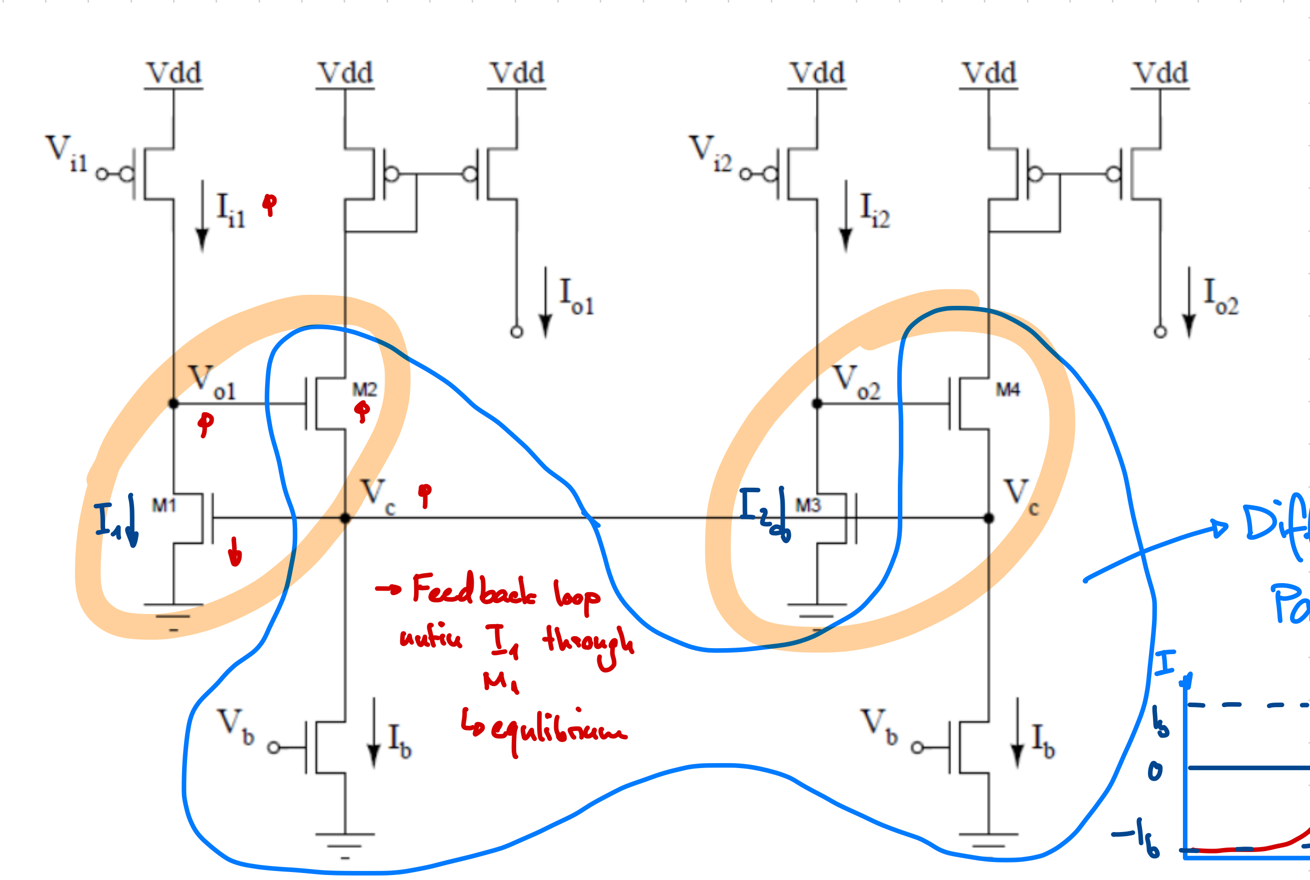
→ Cascode separates Vp of Vd by just putting a transistor in between



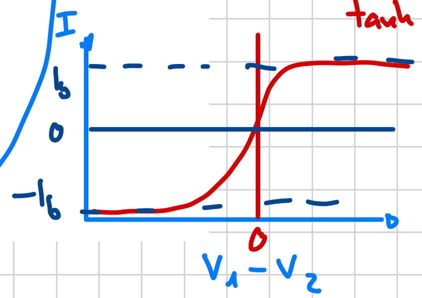
Event
A
SF
G
on
o
o
off
→ Missmatch

WTA

→ Current Correlators



→ Diff. Pair



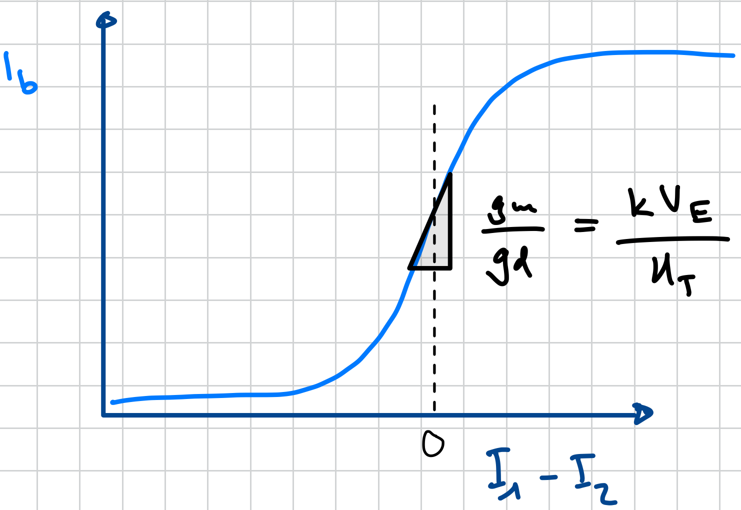
$$V_C \propto \ln\left(\frac{I_1}{I_0}\right) = \ln\left(\frac{I_2}{I_0}\right) \quad \text{if} \quad I_1 = I_2$$

$$\rightarrow I_{o1} = I_{o2} = \frac{I_b}{2} \qquad (\text{if } V_{o1} = V_{o2})$$

$$\hookrightarrow I_{oi} = \frac{I_b}{N}$$

↳ normally:

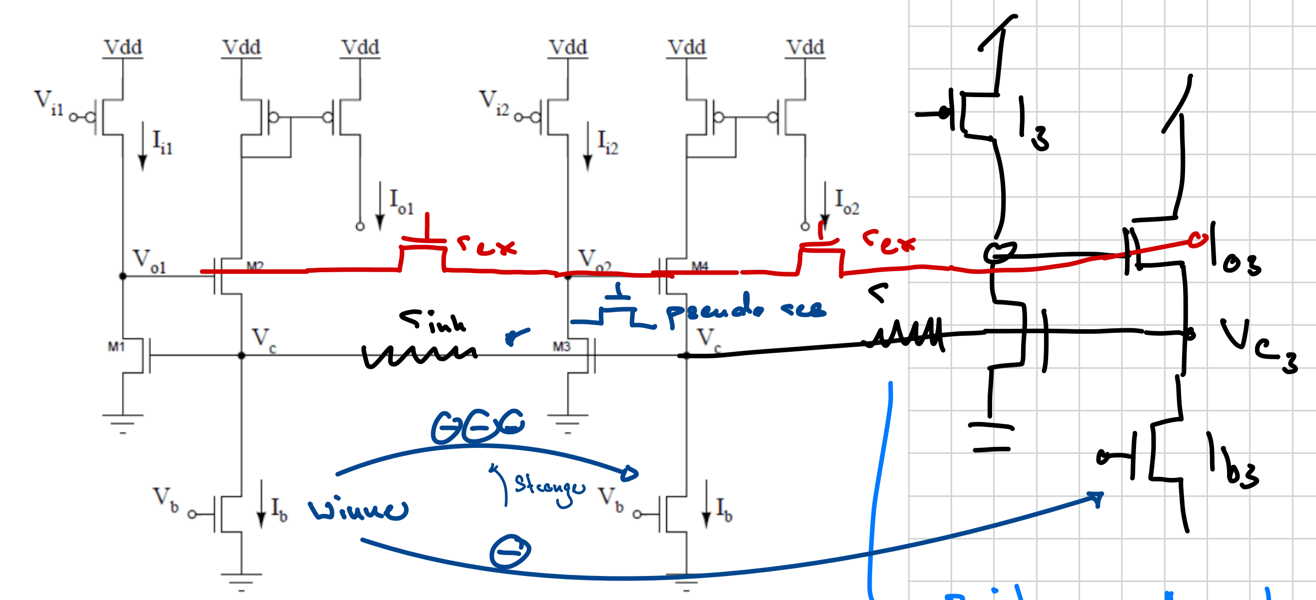
$$V_C = \max \ln\left(\frac{I_i}{I_0}\right)$$



$$\frac{g_m}{g_d} = \frac{k V_E}{U_T}$$

→ Selectivity Gain
↳ speed of picking a winner

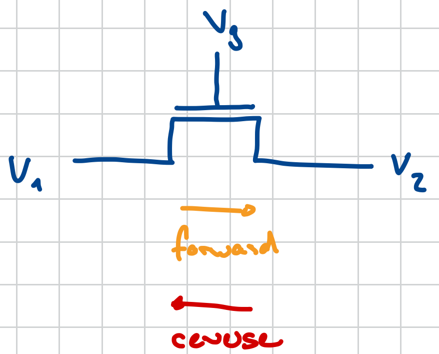
↳ determined by V_E



→ Resistors allow to modulate between WTA and "Everyone wins"

→ Circuit influences more with less resistors in between

↳ Local contest → Local Inhibition



Subthreshold: $I_d = \underbrace{I_0 e^{\kappa V_g - V_2}}_{\text{Forward}} - \underbrace{I_0 e^{\kappa V_g - V_1}}_{\text{Reverse}}$

↳ we want $I = G^* (V_1^* - V_2^*)$ → pseudo conductance

$$I_d = \underbrace{I_0 e^{\kappa V_g}}_{G^*} \left(\underbrace{e^{-V_2}}_{V_2^*} - \underbrace{e^{-V_1}}_{V_1^*} \right)$$

→ conductance

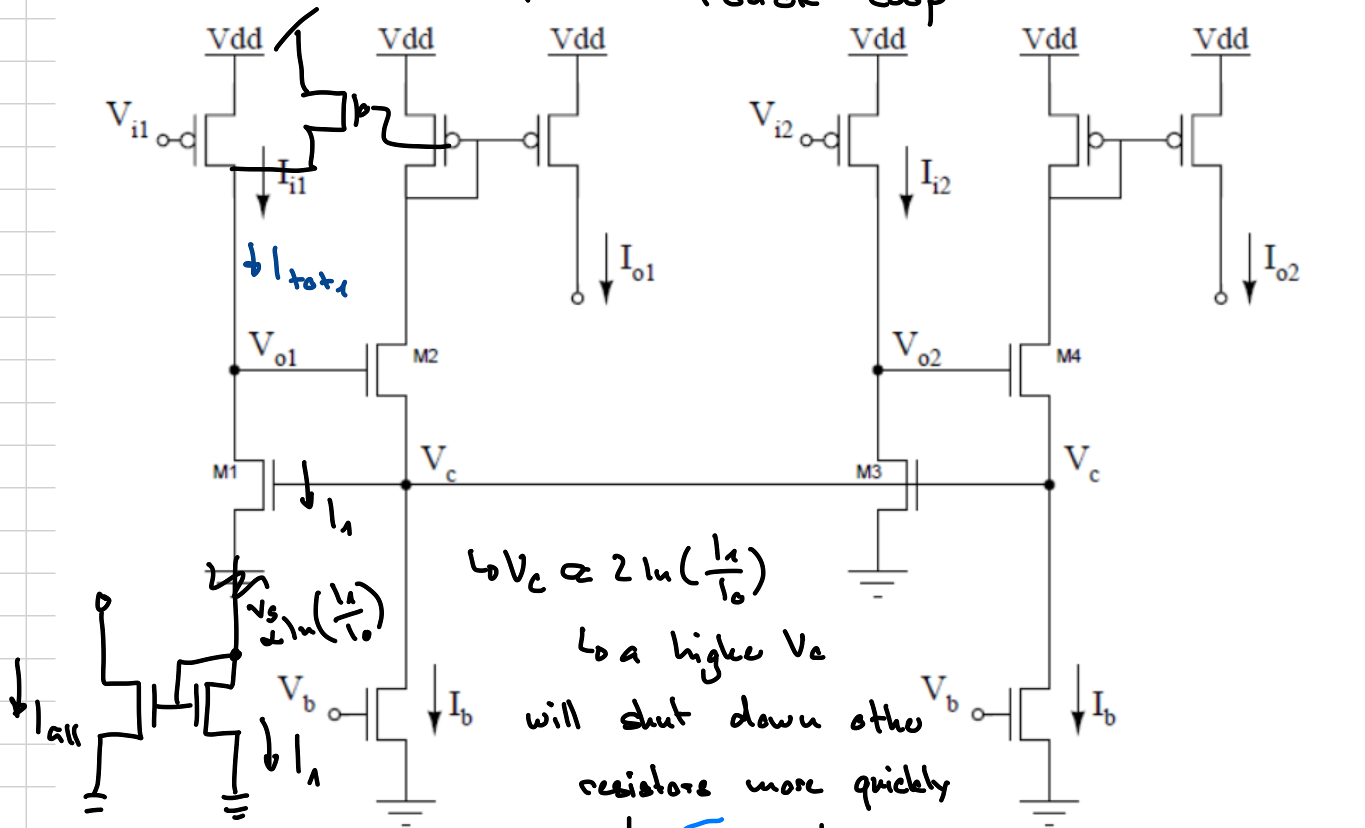
V_{o1}, V_{o2}, V_{o3} connected

↳ Everyone wins if $r = 0$

↳ absolut WTA if $r = \infty$

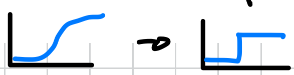
↳ acts as exitatory resitor

→ Positiv Feedback Loop

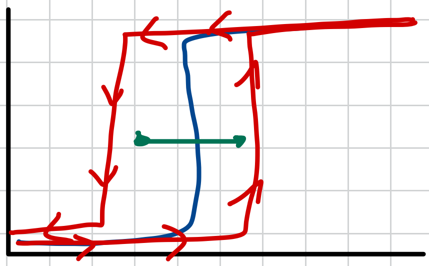


↳ $V_c \propto 2 \ln\left(\frac{I_1}{I_0}\right)$

↳ a higher V_c will shut down other resistors more quickly



Hysteresis



→ Winner can win more easily again

→ State maintaining Machine

↳ Controllable over $\frac{W}{L}$ of PFET

Diode Source regeneration

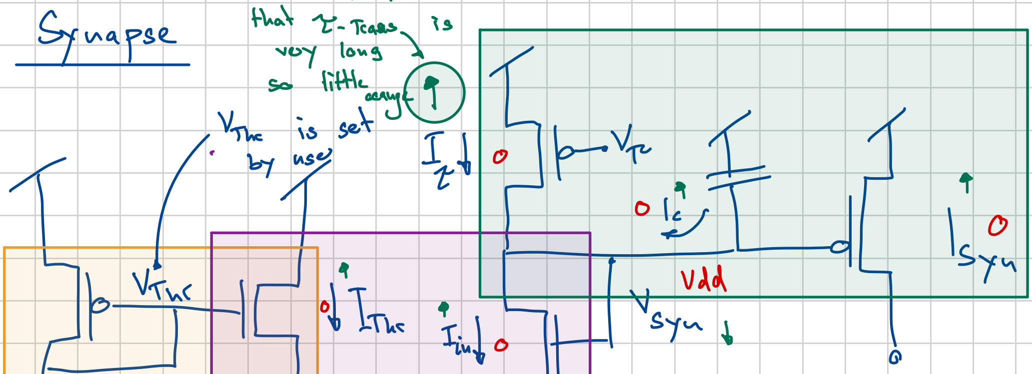
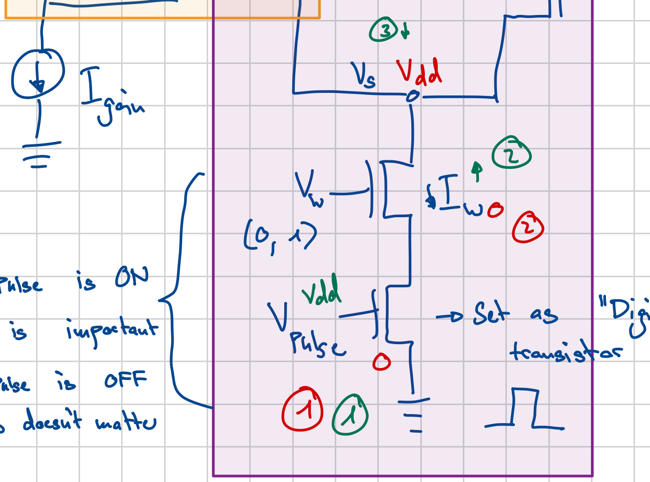
→ To increase the selectivity of circuit

 steeper slope → increase in selectivity Gain

Synapse

We would assume that τ-Trans is very long so little change

V_{Thr} is set by user

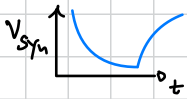



If V_{Pulse} is ON → V_w is important

If V_{Pulse} is OFF → V_w doesn't matter

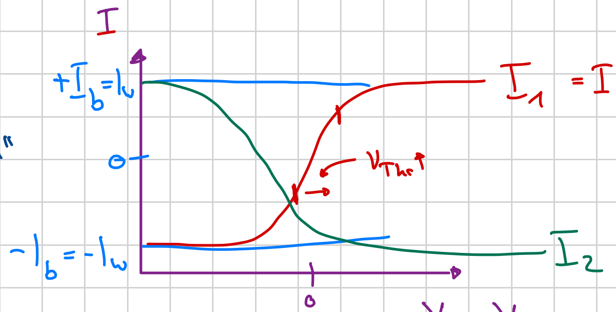
PFET

↳ Gate up → I down



→ RC - Circuit

→ Differential Pair



→ $V_{syn} - V_{Thr}$

A change in V_{Thr} will affect I_{in}

→ Weird Current Mirror since PFET / NFET

↳ Inverse Relationship between I_{Thr} & I_{gain}

When V_{Pulse} = Vdd :

$$\rightarrow I_{Thr} + I_{in} = I_w$$

$$\rightarrow I_w = \underbrace{I_0\, e^{kV_{Th} - V_s}}_{I_{Thr}} + \underbrace{I_0\, e^{kV_{syn} - V_s}}_{I_{in}}$$

(1) $$I_0\, e^{-V_s} = \frac{I_w}{e^{kV_{Th}} + e^{kV_{syn}}}$$

$$\rightarrow I_{in} = I_\tau + I_c$$

$$\rightarrow I_0\, e^{kV_{syn} - V_s} = I_\tau + I_c$$

(2) $$\rightarrow I_w \frac{e^{kV_{syn}}}{e^{kV_{syn}} + e^{kV_{Thr}}} = I_\tau + I_c$$

constant → if derived → 0

(3) $$\rightarrow I_c = C\,\frac{\partial(Vdd - V_{syn})}{\partial t} = -C\,\frac{\partial V_{syn}}{\partial t}$$

$$\rightarrow I_{syn} = I_0\, e^{-kV_{syn} + Vdd}$$

chain Rule

$$\rightarrow \frac{\partial I_{syn}}{\partial t} = \frac{\partial I_{syn}}{\partial V_{syn}} \frac{\partial V_{syn}}{\partial t}$$

$$\frac{\partial I_{syn}}{\partial t} = -k I_{syn} \frac{\partial V_{syn}}{\partial t}$$

(4) $$\boxed{-\frac{\partial V_{syn}}{\partial t} = \frac{1}{k I_{syn}} \frac{\partial I_{syn}}{\partial t}}$$

$$\rightarrow I_w \frac{1}{1 + e^{kV_{Thr}} \cdot e^{-kV_{syn}}} = I_\tau + I_c$$

divide by $e^{kV_{syn}}$

$$\rightarrow I_w \frac{1}{1 + \frac{e^{-kV_{syn}}}{e^{-kV_{Thr}}}} = I_\tau + I_c$$

$$\rightarrow I_w \frac{1}{1+\dfrac{I_{syn}}{I_0 e^{-\kappa V_{Thr}+V_{dd}}}}$$

5

$$\rightarrow I_w \frac{1}{1+\dfrac{I_{syn}}{I_{gain}}}$$

6

$$I_w \frac{1}{1+\dfrac{I_{syn}}{I_{gain}}} = I_\tau + \frac{C}{k I_{syn}} \frac{\partial I_{syn}}{\partial t}$$

divide by I_τ

multiply by I_{syn}

$$\rightarrow \frac{I_w}{I_\tau} \frac{I_{syn}}{\underbrace{1+}_{\text{can be ignored}}\dfrac{I_{syn}}{I_{gain}}} = I_{syn} + \frac{C}{k I_\tau} \frac{\partial I_{gain}}{\partial t}$$

If $I_w >> I_\tau$

$$\rightarrow \boxed{\frac{I_w I_{gain}}{I_\tau} = I_{syn} + \tau \frac{\partial I_{gain}}{\partial t}}$$

↳ Lowpass Filter